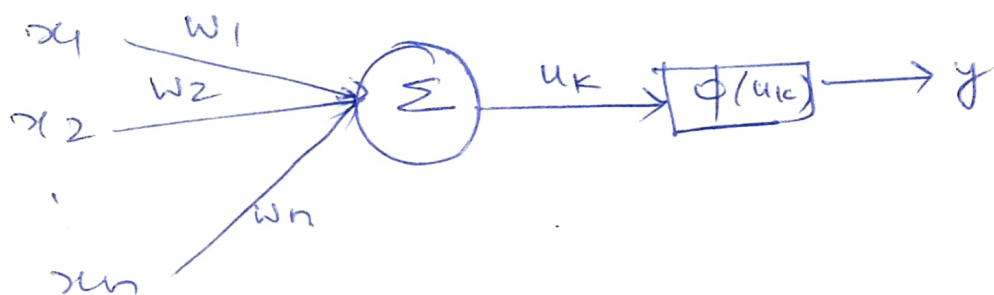


Perceptron:

①

- Frank Rosenblatt, an American psychologist, proposed classical perceptron model (1958)



- A more general computational model of McCulloch-Pitts Neuron Model

Differences:

1. Inputs no longer limited to Boolean values.
2. Introduction of Numerical weights for Inputs and a mechanism for learning weights.

- Refined by Minsky and Papert (1969) referred to as perceptron model

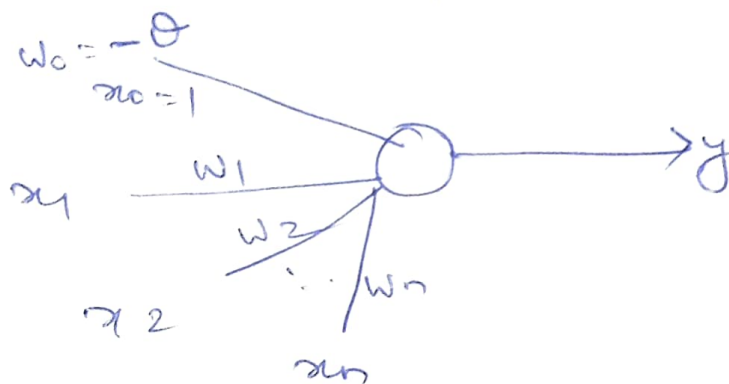
$$y = 1 \quad \text{if} \quad \sum_{i=1}^n w_i x_i \geq \theta$$
$$= 0 \quad \text{if} \quad \sum_{i=1}^n w_i x_i < \theta$$

Rewriting

(2)

$$y = 1 \text{ if } \sum_{i=1}^n w_i x_i \geq 0$$

$$= 0 \text{ if } \sum_{i=1}^n w_i x_i - \theta < 0$$



$$y = 1 \text{ if } \sum_{i=0}^n w_i x_i \geq 0$$

where $x_0 = 1$ and $w_0 = -\theta$

$$\theta = \text{Bias } (w_0)$$

$$= 0 \text{ if } \sum_{i=0}^n w_i x_i < 0$$

Finally
McCulloch Pitts Neuron

$$y = 1 \text{ if } \sum_{i=0}^n x_i \geq 0$$

$$= 0 \text{ if } \sum_{i=0}^n x_i < 0$$

Perceptron,

$$y = 1 \text{ if } \sum_{i=0}^n w_i x_i \geq 0$$

$$= 0 \text{ if } \sum_{i=0}^n w_i x_i < 0$$

Consider a OR function

(3)

x_1	x_2
0	0
0	1
1	0
1	1

OR.

0
1
1
1

$$w_0 + \sum_{i=1}^2 w_i x_i < 0$$

$$\left. \begin{array}{l} 0 \\ 1 \\ 1 \\ 1 \end{array} \right\} w_0 + \sum_{i=1}^2 w_i x_i \geq 0$$

Ist Instance

$$w_0 + w_1 \cdot 0 + w_2 \cdot 0 < 0 \Rightarrow w_0 < 0$$

IInd Instance

$$w_0 + w_1 \cdot 0 + w_2 \cdot 1 \geq 0 \Rightarrow w_2 \geq -w_0$$

IIIrd Instance

$$w_0 + w_1 \cdot 1 + w_2 \cdot 0 \geq 0 \Rightarrow w_1 \geq -w_0$$

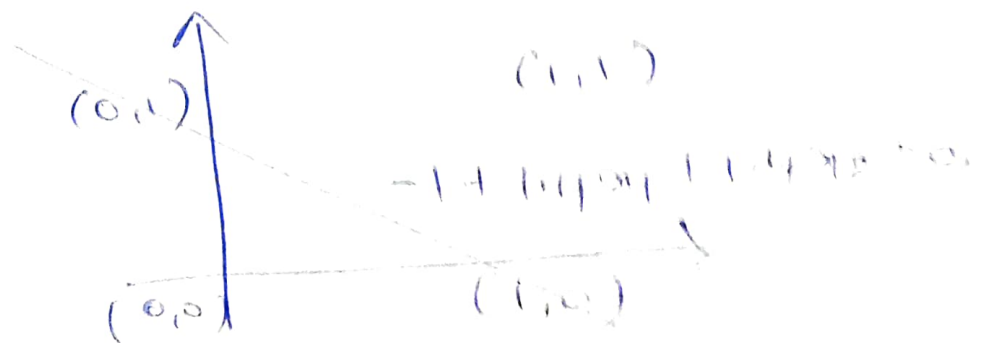
IVth Instance

$$w_0 + w_1 \cdot 1 + w_2 \cdot 1 \geq 0 \Rightarrow w_1 + w_2 \geq -w_0$$

Possible solutions are

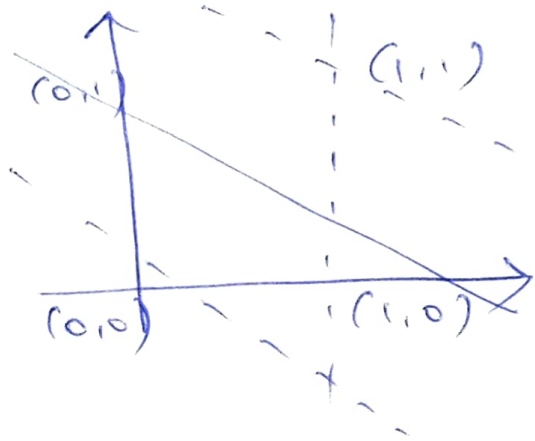
$$w_0 = -1 \quad w_1 = 1.4 \quad w_2 = 1.4$$

Various other solutions are possible



Error function:

Each coin may have different bisection of 2-D plane as like dashed lines.



Say example:

w_1	w_2	errors (misclassification)
-1	-1	3
1.5	0	1
0.45	0.45	3